IVR Immigration AI Receptionist

End-to-end system architecture & data flow — from caller's first word to database record

FastAPI 3000 · GPT-4o · Deepgram nova-3 · ElevenLabs Flash v2.5 · Twilio Media Streams · Redis + PostgreSQL · GoHighLevel CRM

Max 10 concurrent calls · EN/ES bilingual · 8-phase conversation FSM · 12 DB tables

1 INBOUND CHANNELS — HOW CLIENTS REACH THE FIRM

Phone Call
Primary channel · Twilio PSTN

SMS
Text messages · GPT reply

WhatsApp
Up to 4096 chars / msg

Facebook
Messenger · up to 2000 chars

Instagram
DM webhook integration

Web Chat
Website widget embed

2 TWILIO GATEWAY — SECURE ENTRY POINT & CALL AUTHENTICATION

Twilio Webhook Handler · `POST /twilio/voice`

Every inbound call triggers this webhook. Twilio streams real-time audio over a persistent WebSocket connection.

Security first: Every request is verified with **HMAC-SHA1 signature** against your Twilio auth token. Invalid requests are blocked immediately (HTTP 403) before any processing.

- Signature validated
- Concurrency tracked
- Status callbacks
- Recording + Voicemail

3 CALL ROUTING DECISION — SMART ROUTING BASED ON OFFICE HOURS, HOLIDAYS & CAPACITY

OFFICE OPEN + AVAILABLE

AI Voice Agent

- Monday–Friday office hours
- Not a US federal holiday
- Under 10 concurrent calls
- Caller lookup in GHL CRM
- TwiML `<Stream>` → WebSocket pipeline
- Returning callers greeted by name

AT CAPACITY (10 calls)

DTMF IVR Menu

- Press 1 — New consultation
- Press 2 — Case status update
- Press 0 — Front desk (transfer)
- Callback request queue
- Background worker dials back

AFTER HOURS / HOLIDAY

After-Hours & Voicemail

- After-hours greeting played
- Voicemail recording offered
- Transcription via GPT-4o
- Emergency? → on-call attorney
- Follow-up SMS next morning

Primary path: AI Agent via WebSocket

4 REAL-TIME AI VOICE PIPELINE — THE CORE ENGINE (RUNS FOR EVERY SPOKEN WORD)

⚡ SUB-SECOND LATENCY LOOP — TWILIO WEBSOCKET AT `/WS/CALL`

Every audio packet from the caller flows LEFT → CENTER → RIGHT and the voice response streams back in real-time

🎙 AUDIO IN — CALLER'S VOICE

☁ Twilio Media Stream

Format: **mulaw 8kHz**
Encoding: **Base64** over WebSocket JSON
~20ms audio chunks, continuously streamed

🔄 Audio Conversion

Base64 decode → raw bytes
mulaw → linear16 PCM
8kHz → 16kHz upsample (scipy)
Runs in asyncio thread pool

🎤 Deepgram STT

Model: nova-3 (lowest latency)
Language: multi (EN + ES auto)
Endpointing: 200ms silence
VAD Events: SpeechStarted ✓
smart_format · punctuation · paragraphs

⚡ Barge-in Detection

When caller speaks mid-response, `SpeechStarted` event immediately aborts the TTS stream — caller is never talked over

transcript ↓

🧠 AI CORE — INTELLIGENCE ENGINE

📦 Conversation State (Redis)

Key: `conversation:{call_sid}`
TTL: **48 hours**
Stores: phase, language, turns, 18+ intake fields, urgency score, lead score
Caller lookup: Redis cache → GHL CRM

GREETING | ID | TRIAGE | INTAKE | PITCH | BOOKING | CONFIRM | CLOSE

🤖 GPT-4o (OpenAI)

Streaming completions (tokens flow live)
System prompt: **Sofia** (EN/ES, 1024+ tokens)
Prompt caching: prefix cached across all calls
Max tokens per phase: 75–250 (keeps replies short)
Context: last 6 turns + GPT-4o summary of older turns
Intent signals embedded in output text

📋 Intake Engine

18+ structured intake questions
Smart skipping: pre-filled fields auto-skipped
Conditional: questions gated by urgency level
Full name · DOB · immigration status
Case type · court dates · prior history
GPT-4o extracts fields from free speech post-call

⚡ Urgency Classifier ASYNC — runs while TTS plays

GPT-4o function calling · Score 0–10
9–10 EMERGENCY: detained / deportation → transfer now
6–8 HIGH: court in <30 days → callback callback
3–5 MEDIUM: active case → expedite consult
0–2 ROUTINE: informational → standard intake

tokens ↓

🔊 AUDIO OUT — AI VOICE RESPONSE

✂ Sentence Chunker

Buffers streaming GPT tokens
Flushes on: `.!?¿¡` + min 20 chars
Minimizes first-audio latency to caller

🔊 ElevenLabs TTS

Model: Flash v2.5 (ultra-low latency)
EN Voice: Sarah (EXAVITQu4vr4...)
ES Voice: Matilda (XrExE9yKIg1W...)
Output: ulaw_8000 — no re-encoding!
Latency mode: optimize level 3
WebSocket stream-input API

☁ Back to Twilio

mulaw chunks → Base64 encode
WebSocket JSON outbound message
Twilio decodes and plays to caller
~200–400ms end-to-end latency

5 INTENT SIGNALS & REAL-TIME ACTIONS — GPT EMBEDS THESE CODES IN ITS RESPONSES

Emergency Transfer

`EMERGENCY_TRANSFER`

AI detects imminent danger (detained family, deportation order, DV). Instantly warm-transfers to on-call attorney phone number. Urgency alert written to DB.

Book Consultation

`SCHEDULE_NOW`

Caller is ready to book. Pulls available slots from GHL Calendar (Redis cached 1h). Books in GHL as authoritative + Google Calendar for attorney visibility. Sends SMS confirmation.

Language Switch

`LANGUAGE_SWITCH_ES / EN`

Caller switches language mid-call. Instantly swaps voice (Sarah↔Matilda) and system prompt (EN↔ES). Deepgram continues to auto-detect. Seamless bilingual experience.

End Call

`END_CALL`

Graceful close triggered by AI when conversation is complete. Plays farewell message, drains TTS buffer, closes Deepgram connection, triggers post-call pipeline.

6 POST-CALL PROCESSING — RUNS IMMEDIATELY AFTER THE CALL ENDS

Lead Scorer

GPT-4o scores lead 0–100 across 4 dimensions:

Case Value /25 · Urgency /25 · Readiness /25 · Completeness /25

Routing: senior / associate / paralegal attorney

Sentiment Scorer

GPT-4o scores call sentiment −1.0 → +1.0. Detects caller frustration level and confidence flags. Triggers escalation review if score < −0.5.

Structured Intake Extraction

GPT-4o function-calling pass over the full transcript. Normalises free-speech answers into typed fields (dates, enums, booleans). Writes to `immigration_intake` table.

CRM Sync & Follow-Up

Pushes all intake fields to GHL custom fields. Tags contact with case type + urgency level. Enqueues SMS follow-up. Cost tracked (Deepgram + OpenAI + ElevenLabs).

7 DATA STORAGE — THREE-LAYER PERSISTENCE (LIVE → QUEUED → DURABLE)

Redis — Live State & Queues

- `conversation:{call_sid}` — live call state (48h)
- `ghl:phone:{phone}` — CRM cache (24h)
- `slot_cache:{date}` — calendar slots (1h)
- `dash:session:{token}` — dashboard auth (8h)
- `lead_score:{call_sid}` — score cache (30m)

8 async write queues →

db_sync_queue · lead_score_queue
sms_queue · voicemail_queue
analytics_queue · audit_queue
alerts_queue · follow_up_queue

→ drains

DB Worker — Background Drainer

- **Always running** background asyncio loop
- Round-robin across all 8 queues
- BLPOP with 100ms polling interval
- Failed items → `dlq:{queue}` dead-letter
- Writes via Supabase REST client
- Upsert semantics (safe re-delivery)

Cost tracker: Deepgram $0.0043/min
OpenAI $2.50/$10 / 1M tokens
ElevenLabs $0.30 / 1K chars

→ writes

PostgreSQL / Supabase — 12 Tables

clients · conversations · immigration_intake · conversation_messages · call_logs · lead_scores · appointments · urgency_alerts · document_checklists · referral_tracking · voicemails · callback_queue

- DOB & A-numbers encrypted at rest (pgcrypto)
- UUID primary keys throughout
- All timestamps UTC with timezone
- pgvector + uuid-ossp extensions

⇄ sync

GoHighLevel CRM

- Contact create / update per caller
- Tags: case type + urgency level
- Appointments (authoritative copy)
- Custom fields: all 18 intake answers
- GHL webhook events → `/ghl/*`
- Rate limit: 100 req/min (token bucket)
- SMS reminders via GHL pipeline

8 OUTPUTS & REPORTING — WHAT THE FIRM SEES AND ACTS ON

Analytics Dashboard
Call volume · Bookings · Outcomes · Case types · Lead scores · Client pipeline — live at /dashboard/

Intake Records
Per-caller structured data — case type, urgency, court date, immigration status, completeness %

Urgency Alerts
EMERGENCY & HIGH cases flagged instantly. On-call attorney notified. Alert record in DB for audit trail.

SMS / WhatsApp
Booking confirmations · Appointment reminders · After-hours follow-ups · Multilingual (EN/ES)

Consultations Booked
GHL Calendar (authoritative) + Google Calendar (attorney visibility). Dual-write with graceful fallback.

Cost & ROI Tracking
Per-call AI cost (Deepgram + OpenAI + ElevenLabs). Lead score → attorney tier routing for revenue optimisation.

Built-in Resilience & Fallbacks
Circuit Breaker — 3 failures in 60s trips the circuit (30s timeout). Recovers automatically. Pre-recorded filler audio (*"one moment please"*) plays instantly during outages — no dead air. · **Retry with backoff** — transient HTTP errors retried up to 2× with exponential backoff. · **Silence detection** — 15s warning "Are you still there?", 30s hard timeout closes call gracefully. · **SIGTERM grace period** — 30s drain of active WebSocket calls before shutdown.

- ✅ Deepgram
- ✅ ElevenLabs
- ✅ OpenAI
- ✅ GHL CRM

LEGEND Twilio / Communication · Deepgram STT · OpenAI / GPT-4o · ElevenLabs TTS · Redis Cache / Queues · PostgreSQL / Supabase · GoHighLevel CRM · Dashboard / Outputs · Urgency / Alerts

IVR Immigration AI Receptionist · Architecture Diagram · March 2026